

PYTHON STUDENT WORKBOOK

30 Real - World Programming Exercises (Beginner → Advanced)

This workbook is designed to help you practise Python programming using real - life scenarios. For each question:

- Carefully read the scenario
- Understand the task requirements
- Think about the Python concepts you should use
- Write and test your program

NOTE: Do NOT copy solutions. Focus on logical thinking and problem solving.

Exercise 1: Even or Odd Analyzer

A small mobile app wants to help students quickly determine whether numbers are even or odd. Write a Python program that asks the user to enter an integer and displays whether the number is EVEN or ODD. Your program should handle any positive whole number.

Programming Hint: Hint: Use input(), int(), modulus operator %, and an if - else statement.

Exercise 2: Age Classification System

A community registration system needs to classify people based on their age group. Write a program that asks the user to enter their age and then displays: Minor (below 18), Adult (18 – 59), or Senior (60 and above).

Programming Hint: Hint: Use multiple conditional checks (if - elif - else).

Exercise 3: Weather Temperature Reporter

A weather station records daily temperature readings. Create a program that accepts a temperature value and displays: HOT if temperature is above 35 ° C, COLD if below 20 ° C, and NORMAL otherwise.

Programming Hint: Hint: Use comparison operators like > and <.

Exercise 4: Basic Login System

A simple school portal requires users to log in. Store a predefined username and password inside your program. Ask the user to enter login details and display whether login is successful or not.

Programming Hint: Hint: Use logical operator AND to compare credentials.

Exercise 5: Arithmetic Calculator

Design a basic calculator that accepts two numbers and a mathematical operator (+, -, *, /). The program should perform the selected operation and display the result.

Programming Hint: Hint: Use conditional branching based on operator choice.

Exercise 6: Multiplication Table Generator

Teachers often need multiplication tables for classroom activities. Write a program that asks for a number and displays its multiplication table from 1 to 12.

Programming Hint: Hint: Use a for loop with range().

Exercise 7: Sum Accumulator

A mathematics learner wants to find the sum of numbers from 1 up to a given number. Write a program that asks for N and calculates the total sum.

Programming Hint: Hint: Use an accumulator variable inside a loop.

Exercise 8: Countdown Simulator

A rocket launch simulation needs a countdown feature. Write a program that displays numbers from 10 down to 1.

Programming Hint: Hint: Use a while loop and decrement the counter.

Exercise 9: Student Name Storage

A teacher wants to store and display a list of student names. Create a list containing five names and print them one by one.

Programming Hint: Hint: Iterate through a list using a loop.

Exercise 10: GPS Coordinate Viewer

A delivery company stores fixed GPS coordinates of its office. Store latitude and longitude using a tuple and display both values clearly.

Programming Hint: Hint: Access tuple elements using indexing.

Exercise 11: Student Grading Program

An examination system automatically assigns grades. Write a program that accepts marks and displays grade A, B, C, D or F.

Programming Hint: Hint: Use chained conditions.

Exercise 12: ATM Withdrawal Decision

A bank ATM must check if customers have sufficient funds. Assume a fixed balance. Ask the user to enter withdrawal amount and update balance if possible.

Programming Hint: Hint: Compare withdrawal amount with available balance.

Exercise 13: Shopping Basket Totalizer

A supermarket cashier needs to compute the total cost of purchased items. Ask how many items the customer bought and sum their prices.

Programming Hint: Hint: Loop through number of items and keep adding prices.

Exercise 14: Number Guessing Challenge

Create a simple game where the computer stores a secret number. The user keeps guessing until the correct number is entered.

Programming Hint: Hint: Use a loop that continues until condition is satisfied.

Exercise 15: Prime Number Validator

Mathematics students often test whether numbers are prime. Write a program that checks if a number has any divisors other than 1 and itself.

Programming Hint: Hint: Use loop and break statement.

Exercise 16: Employee Bonus Calculator

A company calculates employee bonus using a reusable function. Create a function that returns bonus percentage based on salary.

Programming Hint: Hint: Define a function and use return statement.

Exercise 17: Largest Score Finder

Given a list of scores, determine the highest score achieved. Write logic that compares values to find the largest number.

Programming Hint: Hint: Track maximum value during iteration.

Exercise 18: Attendance Analyzer

A class register stores attendance as P (Present) or A (Absent). Count and display total number of Present students.

Programming Hint: Hint: Use a counter variable.

Exercise 19: Product Price Search Tool

A shopkeeper stores product prices in a dictionary. Allow the user to search for an item and display its price.

Programming Hint: Hint: Check membership using 'in' keyword.

Exercise 20: Secure Password Retry

A system allows only three login attempts. Write a program that blocks access after three incorrect passwords.

Programming Hint: Hint: Use loop with attempt limit.

Exercise 21: Bus Seat Reservation Manager

A transport company books seats until the bus is full. Simulate continuous booking until all seats are taken.

Programming Hint: Hint: Reduce seat count inside a while loop.

Exercise 22: Transaction Balance Calculator

Bank transactions are stored in a list as deposits and withdrawals. Calculate the final account balance after processing all transactions.

Programming Hint: Hint: Add each transaction value to balance.

Exercise 23: Voter Eligibility Filter

Given a list of ages, display only those eligible to vote (18 and above).

Programming Hint: Hint: Use conditional filtering during iteration.

Exercise 24: Marks Average Processor

Create a function that receives a list of marks and returns the average.

Programming Hint: Hint: Use sum() and len().

Exercise 25: Restaurant Ordering System

Customers can order multiple food items from a menu stored as a dictionary. Calculate the total bill after ordering is completed.

Programming Hint: Hint: Use loop until user decides to stop.

Exercise 26: Sentence Vowel Counter

Analyse a sentence and count how many vowels it contains.

Programming Hint: Hint: Convert text to lowercase and iterate characters.

Exercise 27: Inventory Stock Updater

Maintain stock levels of products in a dictionary. Allow user to increase quantity of a selected product.

Programming Hint: Hint: Update dictionary values.

Exercise 28: Student Information Reporter

Store student details (name, age, grade) in a dictionary and display a formatted report.

Programming Hint: Hint: Use dictionary items() method.

Exercise 29: Daily Expense Monitor

Users enter daily expenses. Display total spending and the highest expense.

Programming Hint: Hint: Store values in list and use max().

Exercise 30: Menu - Driven ATM Simulation

Design a continuous ATM program where users can check balance, deposit, withdraw or exit.

Programming Hint: Hint: Use an infinite loop with multiple conditional branches.